41107 MARINE AND OCEAN ENGINEERING

COMPLETE FORMULA REFERENCE — ALL SECTIONS

DTU — Technical University of Denmark | Including all flow-regime thresholds & vortex-pair tables

1. STEADY FLOW AROUND CYLINDERS

1.1 Reynolds Number & Flow Regimes

$Re = Uc \cdot D / \nu$

Uc = free-stream velocity [m/s], D = cylinder diameter [m], ν = kinematic viscosity [m²/s]

Default: $\nu = 1\times10^{-6}$ m²/s (water at ~20°C), ρ = 1025 kg/m³ (seawater)

SMOOTH CYLINDER — regime boundaries:

Regime	Re range	CD (approx)	Strouhal St	Wake character
Sub-critical	$Re < 3\times10^5$	CD ≈ 1.2	St ≈ 0.20	Laminar BL, turbulent wake, regular Kármán shedding
Trans-critical	$3\times10^5 < Re < 3\times10^6$	CD ≈ 0.3–0.9	Irregular	Turbulent BL transition, "drag crisis", chaotic shedding
Super-critical	$Re > 3\times10^6$	CD ≈ 0.6	St ≈ 0.25–0.35	Fully turbulent BL, narrower wake, organised shedding again

⚑ Drag crisis = abrupt CD drop at $Re \sim 3\times10^5$ when laminar BL transitions to turbulent, causing separation point to move downstream.

1.2 Drag Force & Pressure

```
FD = ½ · ρ · CD · D · Uc²          [N/m — per unit length of cylinder]
cp = (p − p∞) / (½·ρ·Uc²)          [pressure coefficient]
cp_stagnation = +1   →  p_stag = p∞ + ½·ρ·Uc²
cp_separation point (sub-critical) ≈ −1.2  to  −2.0  (leeward side)
```

⚑ Form drag dominates for $Re > 10^4$ (~90–95% of total). Friction drag only significant at very low Re.

1.3 Vortex Shedding — Strouhal Number

```
St = fv · D / Uc     →    fv = St · Uc / D
f_lift  = fv             (lift oscillates once per shedding period)
f_drag  = 2·fv           (drag oscillates TWICE per shedding period)
Vortices shed per full period = 2  (one from each side, alternating)
```

⚑ St ≈ 0.2 in sub-critical regime. At resonance (lock-in): fv = fn (natural frequency).

1.4 Force Components — Form vs Friction

Re range	Form drag fraction	Friction fraction	Dominant mechanism
Re < 1	~50%	~50%	Stokes creeping flow
$Re \sim 10^2$	~70%	~30%	Attached flow, growing wake
$Re \sim 10^4$	~90%	~10%	Separated flow, large wake
$Re \sim 10^5$ (sub-crit)	~95%	~5%	Turbulent wake, pressure dominates

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2. OSCILLATORY FLOW — KC REGIMES & VORTEX-PAIR THRESHOLDS

2.1 Keulegan-Carpenter Number

$KC = Um \cdot T / D$

Um = maximum orbital velocity amplitude [m/s], T = wave/oscillation period [s], D = diameter [m]

Physical meaning: KC ≈ (fluid excursion amplitude × π) / D → ratio of stroke to diameter

```
Also written: KC = 2π·a / D   where a = Um/(ω) = fluid excursion amplitude
```

⚑ KC governs SEPARATION in oscillatory flow. Even if Re is huge, small KC → no separation.

2.2 Oscillatory Flow Regimes (Re = 1000, smooth cylinder — Sumer & Fredsøe)

KC range	Regime name	Flow description	NL (lift freq./flow freq.)	Pairs per cycle
KC < 1.1	No separation	Purely viscous, fully attached flow	—	0
1.1 < KC < 1.6	Honji instability	3-D Honji vortices along cylinder axis	—	0
1.6 < KC < 4	Symmetric vortex pair	Pair of symmetric attached vortices; no shedding	—	0
4 < KC < 7	Asymmetric attached pair	Vortex symmetry breaks; lift ≠ 0; still attached	1 (approx)	0
7 < KC < 15	Single-pair regime	One vortex pair shed per full period; transverse street	2	1
15 < KC < 24	Double-pair regime	Two vortex pairs shed per cycle, opposite sides	3	2
24 < KC < 32	Three-pairs regime	Three vortex pairs shed per cycle	4	3
32 < KC < 40	Four-pairs regime	Four vortex pairs per cycle	5	4
KC > 40	Higher-pair regimes	Each +8 KC adds one more pair per cycle	n+1	n

⚑ RULE: Each time KC range increases by ~8, one additional vortex pair is added per cycle. Vortex shedding starts at KC ≈ 7. NL = normalized fundamental lift frequency = lift oscillations per flow cycle.

⚑ For large Re (Re > 4000), regime boundaries shift. Consult Sumer & Fredsøe Fig. 3.15–3.16 for the full (Re, KC) regime map.

2.3 NL Table Summary

KC regime	Pairs convecting away per cycle	NL = lift freq / flow freq	Alias
KC < 7	0 (attached or no separation)	1 (or irregular)	Pre-shedding
7 – 15	1 pair	2	Single-pair
15 – 24	2 pairs	3	Double-pair
24 – 32	3 pairs	4	Three-pairs
32 – 40	4 pairs	5	Four-pairs

⚑ EXAM TRAP: NL=3 → double-pair regime (KC 15–24). NL counts lift oscillations, which equals (pairs+1) because one extra oscillation comes from each flow reversal.

2.4 Wave Orbital Velocity at any depth

```
Um(z) = (π·H/T) · cosh(k(z+h)) / sinh(k·h)     [z measured from seabed]
At surface (z=h):  Um = π·H/T  (deep water)
At seabed (z=0):   Um = π·H / (T·sinh(k·h))
Deep water approx: Um_surface = ω · a = (2π/T)·(H/2)
```

Dispersion relation (must be solved iteratively):

```
ω² = g · k · tanh(k·h)      where ω = 2π/T
```

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3. MORISON EQUATION & HYDRODYNAMIC FORCES

3.1 Full Morison Equation (fixed cylinder)

```
F(t) = ½·ρ·CD·D·U|U|  +  ρ·CM·A·(dU/dt)           [N/m]
```

Drag term: ½·ρ·CD·D·U|U| — velocity-dependent, nonlinear

Inertia term: ρ·CM·A·(dU/dt) — acceleration-dependent, linear

A = π·D²/4 = cross-sectional area of cylinder [m²]

CM = Cm + 1 (total inertia coeff)   where Cm = added mass coeff (potential: Cm=1 → CM=2)

```
For MOVING cylinder (velocity ẋ):
```

```
F(t) = ½·ρ·CD·D·(U−ẋ)|U−ẋ|  +  ρ·Cm·A·(U̇−ẍ)  +  ρ·A·U̇
          drag                      hydrodyn. mass          Froude-Krylov
```

3.2 Maximum Forces (U = Um·sin(ωt))

```
FD,max = ½ · ρ · CD · D · Um²                 [N/m]
FI,max = ρ · CM · (π·D²/4) · Um · ω           [N/m],   ω = 2π/T
FK,max = ρ · (π·D²/4) · Um · ω                [N/m]   (Froude-Krylov only, =
FI,max/CM)
```

Key ratio:

```
FI,max / FD,max = (π² · CM) / (KC · CD)
```

FD peaks at ωt=0° (max velocity). FI peaks at ωt=90° (max acceleration). They NEVER peak together.

Regime guide: KC < 20–30 → inertia-dominated (FI >> FD). KC > 20–30 → drag-dominated.

3.3 Force Components: Froude-Krylov vs Inertia vs Drag

Force component	Formula	Physical origin	When dominant
Froude-Krylov FK	ρ·A·(dU/dt)	Undisturbed pressure gradient (hull absent)	Small structures (D/L < 0.2), CM→1
Hydrodynamic mass FHM	ρ·Cm·A·(dU/dt)	Fluid inertia due to body presence	Always present in water
Total inertia FI = FK + FHM	ρ·CM·A·(dU/dt)	Both above combined	KC < 20; small cylinders
Drag FD	½·ρ·CD·D·U|U|	Pressure difference (wake) + friction	KC > 20; large amplitudes

```
Diffraction important when: D/L > 0.2  (use MacCamy-Fuchs theory for large
cylinders)
D/L < 0.2: Morison equation adequate; FK + Morison coefficients sufficient
```

3.4 CD and CM Empirical Formulas (from exam context, 2 < KC < 12)

```
CD = 1.3 + 0.1·(KC − 12)      [valid for 2 < KC < 12, smooth cylinder]
CM = min(2.0 ; 2.0 − 0.044·(KC − 3))     [valid for 2 < KC < 12]
```

For KC < 2 (no separation): CM → 2.0, CD → 0 (viscous only)

For KC > 20 (drag dominated): CD → 1.2 (sub-critical), CM → 1.2–1.5

3.5 Stagnation & Pressure Coefficient

```
p_stagnation = p∞ + ½·ρ·Uc²      (cp = +1 at front stagnation)
cp = (p − p∞) / (½·ρ·Uc²)
cp_theory = 1 − 4·sin²(θ)      [potential flow around cylinder; θ = angle from
stagnation]
cp at θ=90° (top/bottom) = 1 − 4 = −3    (potential flow)
cp at θ=90° (real, sub-crit) ≈ −2.0 to −2.5    (viscous effects reduce
suction)
```

4. VORTEX-INDUCED VIBRATIONS (VIV)

4.1 Natural Frequency — in Air vs in Water

IN AIR (added mass negligible):

```
ωn = √(k / m_struct)     fn = ωn / (2π)    [Hz]
```

IN WATER (added mass must be included):

```
ωn = √(k / (m_struct + m_added))     where  m_added = ρ · Cm · (π·D²/4)
[kg/m]
```

```
For circular cylinder in potential flow: Cm = 1  →  m_added = ρ·π·D²/4
```

⚑ Added mass can dominate for light structures in water. Always use the "in water" formula for marine problems.

4.2 Reduced Velocity

```
Vr = Uc / (fn · D)      [steady current]
Vr = Um / (fn · D)      [oscillatory flow — use MAX velocity]
```

$f_n = \omega_n/(2\pi)$ [Hz] is the natural frequency IN WATER (including added mass)

4.3 Cross-Flow VIV — Lock-in Thresholds (Steady Current)

Parameter	Value	Condition / Notes
Onset of VIV	Vr ≈ 4–5	Shedding frequency approaches fn; vibration begins
Lock-in range (air)	4.75 < Vr < 8	Shedding locks onto fn; maximum amplitudes at Vr~5.5–6.5
Lock-in range (water)	3 < Vr < ~10	Broader range in water due to lower mass ratio effects
Maximum A/D	~0.5 – 1.5	Depends on Ks; peaks near Vr~5–7
Unlock (end of lock-in)	Vr ≈ 7–8	fv jumps back to Strouhal frequency
Strouhal condition	fv = St·Uc/D	St ≈ 0.2 in sub-critical; Vr_lockin ≈ 1/St = 5

4.4 In-Line VIV — Instability Regions (Steady Current)

Region	Vr range	Cause	Max Ax/D
1st instability	~1.25 < Vr < ~2.5	Normal Kármán + symmetric secondary shedding; fx = 3·St·Uc/D → fx=fn at Vr=1/(3·St)≈1.7	~0.05–0.10
2nd instability	~2.5 < Vr < ~3.8	Normal Kármán drag oscillation at 2·fv; 2·fv=fn at Vr=1/(2·St)=2.5	~0.10–0.15
3rd (cross-flow region)	Vr~5–8	Cross-flow VIV drives in-line at 2·fy; much larger Ax due to strong vortices	~0.3–0.5 (when Ay/D > 0.2)

```
Ax/Ay ≈ 1/10  (in-line amplitude is ~10× smaller than cross-flow amplitude)
```

⚑ 1st instability: $V_r = 1/(3\cdot St) \approx 1.7$. 2nd instability: $V_r = 1/(2\cdot St) \approx 2.5$. Cross-flow lock-in: $V_r \approx 1/St \approx 5$. All with $St = 0.2$.

4.5 Stability (Scruton) Parameter & Amplitude Model

```
Ks = 2 · M · δs     where  δs = 2π·ζs  (log. decrement of structural damping)
M  = m_struct / (ρ · π·D²/4)      [mass ratio — non-dim structural mass]
Amplitude model (simplified):   A/D = 1.5 / Ks   (cross-flow, approx)
```

Higher Ks → more damping/heavier → smaller VIV. Key design lever: add structural damping or mass.

Ks value	Expected cross-flow A/D	Design implication
Ks < 1	A/D → 1.0+	Large VIV; suppression devices needed (strakes, fairings)
1 < Ks < 3	A/D ~ 0.3–0.8	Moderate VIV; check fatigue life carefully
3 < Ks < 10	A/D ~ 0.1–0.3	Small VIV; often acceptable
Ks > 10	A/D < 0.1	VIV suppressed; structure is very heavy or highly damped

4.6 Free Decay Test — Damping & Natural Frequency

```
δ = ln(yn / yn+1)                      [logarithmic decrement]
ζ = δ / √(4π² + δ²)                    [damping ratio — exact formula]
ζ ≈ δ / (2π)                            [simple approx; valid for ζ < 0.2]
ωd = 2π / Td                           [damped natural frequency from
measured period]
ωn = ωd / √(1 − ζ²)                    [undamped natural frequency]
```

⚑ yn and yn+1 are SAME-SIGN consecutive peaks. Td = damped period (one full oscillation).

5. REGULAR WAVE THEORY & DISPERSION

5.1 Linear Wave Theory — Key Formulas

$\omega^2 = g \cdot k \cdot \tanh(k \cdot h)$ [dispersion relation — solve iteratively for k]

$L = 2\pi/k$, $T = 2\pi/\omega$, $c = L/T = \omega/k$ [wavelength, period, phase speed]

Deep water ($h > L/2$): $\tanh(kh) \to 1 \to c = gT/(2\pi) \to L_0 = gT^2/(2\pi) \approx 1.56 \cdot T^2$

Shallow water ($h < L/20$): $\tanh(kh) \to kh \to c = \sqrt{(g \cdot h)}$ (non-dispersive)

Quantity	Deep water formula	Shallow water formula	General
Phase speed c	$gT/(2\pi) \approx 1.56T$ [m/s]	$\sqrt{(g \cdot h)}$	ω/k
Group velocity cg	$c/2$	$c = \sqrt{(g \cdot h)}$	$c \cdot n$, $n=½(1+2kh/\sinh(2kh))$
Wavelength L	$gT^2/(2\pi) \approx 1.56T^2$ [m]	$T \cdot \sqrt{(g \cdot h)}$	$2\pi/k$
Max horiz. velocity Um	$\pi H/T = \omega a$ [at surface]	$\pi H/(T \cdot kh) = c \cdot H/(2h)$	$(\pi H/T) \cdot \cosh(k(z+h))/\sinh(kh)$
Max vert. velocity wm	$\pi H/T$ (same as Um)	0 at bottom	$(\pi H/T) \cdot \sinh(k(z+h))/\sinh(kh)$
Max horiz. accel.	$\omega^2 a = (2\pi/T)^2 \cdot (H/2)$	—	$\omega^2 \cdot a \cdot \cosh(k(z+h))/\sinh(kh)$

5.2 Encounter Frequency (Ship Moving in Waves)

$\omega e = \omega - (\omega^2/g) \cdot V \cdot \cos(\beta)$ [encounter angular frequency]

$Te = 2\pi / \omega e$ [encounter period]

β = wave heading angle (DTU convention):

$\beta = 180°$: head seas $\to \cos(180°) = -1 \to \omega e = \omega + (\omega^2/g) \cdot V > \omega$ (encounters faster)

$\beta = 0°$: following $\to \cos(0°) = +1 \to \omega e = \omega - (\omega^2/g) \cdot V < \omega$ (encounters slower)

$\beta = 90°$: beam seas $\to \cos(90°) = 0 \to \omega e = \omega$ (no Doppler shift)

$\beta = 135°$: bow-quartering $\to \cos(135°) = -0.707$

To find absolute T from Tenc: solve quadratic in ω:

$a \cdot \omega^2 - \omega + \omega e = 0$ where $a = -V \cdot \cos(\beta)/g \to \omega = (1 \pm \sqrt{(1-4a \cdot \omega e)}) / (2a)$

1 knot = 0.5144 m/s. For following seas (ωe can be very small or negative) — DTU exams avoid this.

5.3 Wave Breaking Limit (Miche Criterion)

$Hmax = 0.142 \cdot L \cdot \tanh(k \cdot h)$

Deep water limit: $H/L < 1/7 \approx 0.142$

Shallow water limit: $H < 0.78 \cdot h$ (solitary wave breaking)

6. WAVE STATISTICS & SPECTRAL ANALYSIS [EXAM-CRITICAL SECTION]

6.1 Spectral Moments — Definitions

$mn = \int_0^\infty \omega^n \cdot S(\omega) \cdot d\omega$ [n-th spectral moment]

Key moments:

$m_{-1} = \int_0^\infty \omega^{-1} \cdot S(\omega)\, d\omega$ (mean energy period)

$m_0 = \int_0^\infty S(\omega)\, d\omega = \sigma^2$ (variance of sea surface = area under spectrum)

$m_1 = \int_0^\infty \omega \cdot S(\omega)\, d\omega$

$m_2 = \int_0^\infty \omega^2 \cdot S(\omega)\, d\omega$

$m_4 = \int_0^\infty \omega^4 \cdot S(\omega)\, d\omega$

Discrete approximation (numerical integration):

$mn \approx \Sigma_i\, \omega_i^n \cdot S(\omega_i) \cdot \Delta\omega$

UNIT CHECK: $S(\omega)$ in [m²/s/rad] → m_0 in [m²], m_1 in [m²rad/s⁻¹=m²/s·rad], m_2 in [m²rad²/s²]. Always check units!

6.2 Integral Wave Parameters from Spectral Moments

Parameter	Symbol	Formula	Physical meaning
Significant wave height	$Hs = H_{1/3}$	$Hs = 4\cdot\sqrt{m_0}$	Mean of top 1/3 waves; "typical storm wave"
Variance of surface	σ^2	$\sigma^2 = m_0$	Area under wave energy spectrum
RMS surface elevation	σ	$\sigma = \sqrt{m_0}$	Standard deviation of $\eta(t)$
Mean wave period (T_1)	$T_1 = Tm$	$T_1 = 2\pi \cdot m_0/m_1$	Period at mean frequency
Mean energy period	$TE = T_{-1}$	$TE = 2\pi \cdot m_{-1}/m_0$	Weighted by energy; used in power calculations
Zero-crossing period	$Tz = T_2$	$Tz = 2\pi \cdot \sqrt{m_0/m_2}$	Mean time between zero up-crossings
Mean crest period	Tc	$Tc = 2\pi \cdot \sqrt{m_2/m_4}$	Mean time between wave crests
Peak period	Tp	$Tp = 2\pi/\omega p$ (ωp = argmax $S(\omega)$)	Period at spectral peak
Significant period	Ts	$Ts \approx 0.95\cdot Tp$ (Bretschneider)	Mean period of highest 1/3 waves

WARNING: Different textbooks use different symbols! Tz may be called T_0, Tm, T_2. Always check the definition used in the specific problem.

6.3 Period Relationships for Standard Spectra

Spectrum type	Tz/Tp	T_1/Tp	Ts/Tp
Bretschneider	≈ 0.71	≈ 0.77	≈ 0.95
JONSWAP (γ=3.3)	≈ 0.74	≈ 0.78	≈ 0.97
General relation	$Tz < T_1 < TE < Tp$	—	—

6.4 Parameterised Wave Spectra

Bretschneider Spectrum:

$$S(\omega) = (A/\omega^5) \cdot \exp(-B/\omega^4) \qquad [m^2 s/rad]$$
$$A = (Hs^2/4\pi) \cdot (2\pi/Tz)^4$$
$$B = (1/\pi) \cdot (2\pi/Tz)^4$$
Peak frequency: $\omega p = (4B/5)^{(1/4)} = (B/5)^{(1/4)} \cdot 4^{(1/4)}$

JONSWAP Spectrum (North Sea, developing seas):

$$S(\omega) = (\alpha \cdot g^2/\omega^5) \cdot \exp[-(5/4)\cdot(\omega p/\omega)^4] \cdot \gamma^{\exp[-(\omega-\omega p)^2/(2\sigma^2\omega p^2)]}$$
γ = peak enhancement factor (typically 3.3 for North Sea)
$\sigma = 0.07$ for $\omega \le \omega p$; $\sigma = 0.09$ for $\omega > \omega p$

Pierson-Moskowitz Spectrum (fully developed, wind-speed input):

$$S(\omega) = (8.1\times10^{-3} \cdot g^2/\omega^5) \cdot \exp[-0.74\cdot(g/(V\cdot\omega))^4]$$
V = wind speed [m/s] at 19.5 m above surface

$S(\omega)d\omega = S(f)df$ → $S(f) = S(\omega)\cdot 2\pi$ (conversion between angular and temporal frequency)

6.5 Response Spectrum & Transfer Function

$$SR(\omega) = |\Phi(\omega)|^2 \cdot S\zeta(\omega) \qquad \text{[response spectrum]}$$
$$|\Phi(\omega)| = \sqrt{(SR(\omega) / S\zeta(\omega))} \qquad \text{[transfer function magnitude from spectra]}$$

Response spectral moments:

$$mR,n = \int_0^\infty \omega^n \cdot SR(\omega)\, d\omega = \int_0^\infty \omega^n \cdot |\Phi(\omega)|^2 \cdot S\zeta(\omega)\, d\omega$$

Significant response amplitude (e.g. heave):

$$z_{1/3} = 2\cdot\sqrt{mR,0} \qquad \text{[significant amplitude = 2×RMS]}$$

Unit conversion — if $\Phi(\omega)$ in [rad/m], convert to [deg/m]:

$$|\Phi|\_deg/m = |\Phi|\_rad/m \times (180/\pi)$$

$SR\_deg = SR\_rad \times (180/\pi)^2$

⚑ CRITICAL UNIT CHECK: $R(\omega)$ in [rad²s/rad] → $|\Phi|$_rad/m → must multiply by $180/\pi$ for [deg/m]. Always state units when computing or comparing.

6.6 Phase Angle of Transfer Function (Complex TF)

$\Phi(\omega) = Re + i \cdot Im \quad \rightarrow \quad |\Phi| = \sqrt{(Re^2+Im^2)}, \quad \varphi = \text{atan2}(Im, Re)$

Quadrant table for phase angle:

Quadrant	Re sign	Im sign	ϕ formula	ϕ range
Q1	+ (>0)	+ (>0)	ϕ = arctan(Im/Re)	0° – 90°
Q2	− (<0)	+ (>0)	ϕ = 180° − arctan(|Im|/|Re|)	90° – 180°
Q3	− (<0)	− (<0)	ϕ = 180° + arctan(|Im|/|Re|)	180° – 270°
Q4	+ (>0)	− (<0)	ϕ = 360° − arctan(|Im|/|Re|)	270° – 360°

⚑ Phase LAG = angle by which response lags the wave. DTU reports phase as the argument of Φ in [0°, 360°].

6.7 Probability & Statistics of Ocean Waves

Surface elevation distribution (Gaussian):

$p(\eta) = (1/(\sigma\sqrt{(2\pi)})) \cdot \exp(-\eta^2/(2\sigma^2)) \quad \text{where } \sigma = \sqrt{m_0}$

Wave HEIGHT distribution (Rayleigh — narrow-band):

$p(H) = (H/(4m_0)) \cdot \exp(-H^2/(8m_0))$

$P(H > h) = \exp(-h^2/(8m_0)) = \exp(-2 \cdot (h/Hs)^2)$

Wave CREST/AMPLITUDE distribution (Rayleigh):

$P(X > x) = \exp(-x^2/(2m_0))$ [exceedance probability for amplitudes/peaks]

Equivalently: $P(X > x) = \exp(-(x/\sigma)^2/2)$ where $z = x/\sigma$ → use z-table: $P(Z > z)$

Significant wave height from statistics:

$Hs = H_1/_3 = 4 \cdot \sqrt{m_0} = 4 \cdot \sigma$

$H_1/_{10} = 1.27 \cdot Hs$ (mean of top 10%)

$H_1/_{100} = 1.67 \cdot Hs$ (mean of top 1%)

Maximum wave in N waves (Most Probable Largest):

$H\_MPL = Hs \cdot \sqrt{(0.5 \cdot \ln N)}$ [MPL wave height]

$\eta\_MPL = \sqrt{(2 \cdot m_0 \cdot \ln N)}$ [MPL amplitude; Ochi 1973]

$\eta\_MPL = \sigma \cdot \sqrt{(2 \cdot \ln N)}$ (same, using $\sigma = \sqrt{m_0}$)

Number of waves: $N = T\_storm / Tz$

Number of exceedances of level x in duration T:

$Nx = (T/Tz) \cdot \exp(-x^2/(2m_0))$

Standard normal (z-table) key values:

z = x/σ	P(Z > z) [exceedance]	Practical meaning
1.00	15.87%	~16% chance of exceedance per observation
1.28	10.00%	10% exceedance
1.645	5.00%	5% exceedance (1 in 20)
2.00	2.28%	~2% (close to 1 in 50)
2.04	2.06%	σ=√1.5, a=2.5 → z=2.04 → ~2.1% exam example
2.33	1.00%	1% exceedance (1 in 100)
3.00	0.13%	~1 in 800
3.72	0.01%	~1 in 10,000

6.8 Forward Speed — Wave Spectrum Transformation

```
S*(ωe) = S(ω) / |1 − 2ω·V·cos(β)/g|      [encountered spectrum]
```

Applied when ship moves through irregular seas. ωe = ω − (ω²/g)·V·cos(β).

DTU 41107 only considers β ∈ [90°, 270°] (beam to head to beam). Following sea complications excluded.

6.9 Spectral Width Parameter

```
ε² = 1 − m₁²/(m₀·m₂)          [spectral width; ε=0 narrow-band, ε=1 broad-band]
q = 1 − m₁²/(m₀·m₂)           [alternative width, same formula as ε²]
Narrow-band: ε < 0.3.    Broad-band: ε > 0.6
```

Rayleigh distribution for peaks assumes NARROW-BAND spectrum. Wide-band → Rayleigh is approximate.

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7. SHIP MOTION DYNAMICS & TRANSFER FUNCTIONS

7.1 Degrees of Freedom

Motion	Direction	Axis	Restoring?
Surge	Longitudinal (x)	—	No (no hydrostatic restoring)
Sway	Transverse (y)	—	No
Heave	Vertical (z)	—	YES (waterplane area)
Roll	Rotation about x	Longitudinal	YES (transverse stability)
Pitch	Rotation about y	Transverse	YES (longitudinal stability)
Yaw	Rotation about z	Vertical	No

7.2 Heave Natural Frequency (Vertical Cylinder / Simple Body)

```
ωn_heave = √(g/T_draught)                [neglecting added mass; T = draught]
```

With added mass (more accurate):

```
ωn_heave = √(ρ·g·Awl / (m + m_a))
Awl = π·D²/4 (waterplane area of cylinder)
m = ρ·Awl·T  (floating body mass = displaced water mass)
m_a = added mass in heave (depends on geometry)
```

For vertical cylinder: simplified formula gives ωn = √(g/T). Larger draught → lower ωn → better seakeeping.

7.3 Linear Equations of Motion (Single DOF)

```
(m + m_a)·ẍ + b·ẋ + c·x = F_wave(t)     [general 1-DOF ship motion]
m_a = added (hydrodynamic) mass
b   = damping coefficient (radiation + viscous)
c   = hydrostatic restoring coefficient
```

7.4 Response in Regular Waves — Transfer Function (RAO)

```
z(t) = za·cos(ωt + εzζ)    →    za = |Φ(ω)| · ζa    [linear: response proportional to wave amplitude]
|Φ(ω)| = za / ζa           [magnitude of Transfer Function = RAO]
Phase angle: φ = atan2(Im{Φ}, Re{Φ})
```

From complex TF components:

```
|Φ| = √(Re² + Im²)
φ   = atan2(Im, Re)     (four-quadrant; see Section 6.6 table)
```

RAO = Response Amplitude Operator (alternative name for transfer function magnitude).

7.5 Froude-Krylov Assumption

```
Assumption: the presence of the hull has NO effect on the incident waves.
Froude-Krylov exciting force: F_FK = −∫∫ p_incident · n̂ dS
Valid when D/L < 0.2 (slender body; diffraction small).
For D/L > 0.2: diffraction must be included (full panel method).
```

---

8. SHIP HYDROSTATICS & STABILITY

8.1 Buoyancy & Archimedes

```
Δ = ρ · g · V_displaced      [buoyancy force = weight = displacement in Newtons]
Δ_tons = ρ · V / 1000        [displacement in metric tons; ρ in kg/m³, V in m³]
Freshwater: ρ = 1000 kg/m³  →  1 ton = 1 m³
Saltwater:  ρ = 1025 kg/m³  →  1 ton = 0.976 m³
For box barge: V = L·B·T   →   Δ = ρ·L·B·T [kg]
```

8.2 Metacentric Height — Transverse Stability

```
GM_T = KB + BM_T − KG
```

Quantity	Formula (rectangular box)	Physical meaning
KB	T/2	Height of centre of buoyancy above keel
BM_T	$B^2/(12\cdot T)$ = Iwl/V	Metacentric radius; $I_{wl} = L\cdot B^3/12$
KG	Depends on load distribution	Height of centre of gravity above keel
KM_T = KB+BM_T	$T/2 + B^2/(12T)$	Height of metacentre above keel
GM_T	KM_T – KG	Transverse metacentric height

```
GZ = GM_T · sin(φ)      [righting lever — small angle approximation]
GMT = GZ / sin(φ)       [from measured GZ at angle φ]
```

- GM_T > 0: stable upright. GM_T < 0: unstable upright (will find new equilibrium).
- IMO minimum: GMT ≥ 0.15 m for conventional ships.

8.3 Metacentre at Waterplane — Special Condition

```
KM = T  →  KB + BM = T  →  T/2 + B²/(12T) = T  →  T = B/√6
```

This means M lies exactly at the waterline level. $T = B/\sqrt{6} \approx 0.408\cdot B$.

8.4 Stability of Square Log (Classic Result)

For $\rho_{log} = \rho_{water}/2$: T = b/2 (half-submerged).

```
KB = b/4,    BM = b/6,    KG = b/2
GM_T = b/4 + b/6 − b/2 = (3+2−6)b/12 = −b/12 < 0  →  UNSTABLE upright
```

- A square log with density = ρ/2 is ALWAYS unstable in the symmetric upright position. It will rotate to a new stable equilibrium position (e.g. tilted ~45°).

8.5 Buoyancy & Trim (Longitudinal Stability)

```
GMT_L = KM_L − KG  ≈ GML  (longitudinal metacentric height, >>GM_T)
BM_L = IL_wl / V = L²/(12·T)  (for rectangular box)
Change in trim from off-centre weight w at distance d from midship:
t = w · d · L / (Δ · GML)   [total trim change bow-stern]
Change in draught at bow: ta = t · (L/2 + d_fwd) / L
Change in draught at stern: tf = t · (L/2 − d_fwd) / L  (or use ta + tf = t)
```

9. SHIP RESISTANCE & MODEL SCALING

9.1 Non-Dimensional Parameters

$Fn = V / \sqrt{(g \cdot L)}$ [Froude number — governs wave-making resistance]
$Re = V \cdot L / \nu$ [Reynolds number — governs viscous resistance]

Froude similarity (Froude scaling):

$Vm/Vs = 1/\sqrt{\alpha}$ [model speed at corresponding (Froude-equal) speed]
$\alpha = Ls/Lm$ [length scale ratio]

9.2 Scaling Laws (Froude Similarity, same fluid)

Quantity	Scale factor	Example: α=40
Length L	α	40
Speed V	√α	√40 = 6.32
Time T	√α	6.32
Force / Resistance F	α^3	64,000
Mass / Displacement	α^3	64,000
Power P	α^3.5	10,119
Wetted area	α^2	1,600

$r = Fs/Fm = \alpha^3$ [ship-to-model resistance ratio, same fluid]
Model mass: $mm = ms / \alpha^3$ (geometric/Froude-scaled model displacement)

9.3 ITTC-1978 Resistance Decomposition

$CT = CF \cdot (1+k) + \Delta CF + CR + CA + CAAS$
$CF = 0.075 / (\log_{10}(Re) - 2)^2$ [ITTC 1957 friction line]
$CR = CT\_model - CF\_model \cdot (1+k)$ [residuary (wave) resistance; = f(Fn)]
Resistance: $R = ½ \cdot \rho \cdot V^2 \cdot S \cdot CT$ [S = wetted surface area]

10. STRENGTH OF SHIPS — CROSS-SECTION & BENDING

10.1 Neutral Axis

$zn = \Sigma(Ai \cdot zi) / \Sigma(Ai)$ [weighted centroid of cross-section]

For thin-walled sections: Ai = plate_length_i × t_i, zi = centroid height of plate i

⚑ Plates parallel to horizontal axis contribute area at their centroidal height. Vertical plates contribute area at their mid-height.

10.2 Second Moment of Area (Moment of Inertia)

$I = \Sigma(Ii + Ai \cdot ai^2)$ [Parallel axis theorem (Steiner)]
Ii = own second moment of plate i about its own centroid
ai = distance from plate i centroid to overall neutral axis = $|zi - zn|$

For a thin horizontal plate (width b, thickness t, at height z from baseline):

$Ii = b \cdot t^3/12 \approx 0$ (negligible for thin plates) → contribution = $Ai \cdot ai^2$

For a thin vertical plate (height h, thickness t):

$Ii = t \cdot h^3/12$ (NOT negligible for tall plates) → contribution = $t \cdot h^3/12 + Ai \cdot (z\_c-zn)^2$

10.3 Bending Stress (Navier's Formula)

$\sigma = M \cdot y / I$ [bending stress at distance y from NA]

```
Z = I / y_max                      [section modulus; σ_max = M/Z]
```

$y = |z - z_n|$ = distance from neutral axis to the point of interest

Maximum stress occurs at the point furthest from neutral axis (deck or keel).

🚩 For HOGGING (sagging up at ends): keel in tension, deck in compression. For SAGGING (bending down): deck in tension, keel in compression. Tension is typically more critical for fatigue design.

10.4 Load Distribution & Equilibrium

```
Bonjean curves: cross-sectional area vs draught (used to find buoyancy distribution)
Equilibrium: ∫w(x)dx = ∫b(x)dx    (total weight = total buoyancy)
Net load q(x) = w(x) - b(x)        [distributed load = weight minus buoyancy per metre]
Shear force: V(x) = ∫0^x q(ξ)dξ    (integrate net load from bow)
Bending moment: M(x) = ∫0^x V(ξ)dξ  (integrate shear force)
```

🚩 V and M must both = 0 at free ends (bow and stern). This is the boundary condition check.

MASTER QUICK-REFERENCE — ALL KEY FORMULAS AT A GLANCE

A. Dimensionless Numbers

Number	Symbol	Formula	Governs
Reynolds	Re	Uc·D/ν	Viscous vs inertia; flow regime
Strouhal	St	fv·D/Uc ≈ 0.2	Vortex shedding frequency
Keulegan-Carpenter	KC	Um·T/D	Separation in oscillatory flow
Froude	Fn	V/√(gL)	Wave-making (ship)
Reduced velocity	Vr	Uc/(fn·D) or Um/(fn·D)	VIV lock-in proximity
Mass ratio	M	m/(ρπD²/4)	VIV amplitude
Stability param.	Ks	2·M·2π·ζs	VIV suppression

B. Regime Thresholds — At a Glance

Regime type	Threshold	Below →	Above →
Re: Sub→Trans (smooth)	Re = 3×10^5	CD≈1.2, St=0.2, regular Kármán	Drag crisis begins
Re: Trans→Super (smooth)	Re = 3×10^6	Chaotic, no regular shedding	CD≈0.6, St≈0.3, organised again
KC: No sep → Honji	KC = 1.1	No separation (creeping)	3D Honji vortices
KC: Honji → Symmetric pair	KC = 1.6	Honji instability	Symmetric attached pair
KC: Symmetric → Asymmetric	KC = 4	Attached symmetric vortices	Asymmetric; lift≠0
KC: Asymmetric → Single-pair SHEDDING	KC = 7	Attached asymmetric vortices	VORTEX SHEDDING begins (NL=2)
KC: Single → Double pair	KC = 15	Single-pair (NL=2, 1 pair/cycle)	Double-pair (NL=3, 2 pairs/cycle)
KC: Double → Triple pair	KC = 24	Double-pair (NL=3)	Three-pairs (NL=4, 3 pairs/cycle)
KC: Triple → Four pairs	KC = 32	Three-pairs (NL=4)	Four-pairs (NL=5)
Each additional KC +8	Every 8 KC	n pairs/cycle	n+1 pairs/cycle
VIV onset (cross-flow)	Vr ≈ 4–5	No significant VIV	VIV begins; approaches lock-in
VIV lock-in (cross-flow)	Vr ≈ 5–7	Pre-lock-in small amplitudes	Lock-in: fv=fn, large A/D
VIV unlock	Vr ≈ 7–8	Lock-in active	fv jumps back to Strouhal
In-line: 1st instability	Vr ≈ 1.25–2.5	No in-line VIV	1st instability (Ax/D~0.05-0.10)
In-line: 2nd instability	Vr ≈ 2.5–3.8	1st instability or no VIV	2nd instability (Ax/D~0.10-0.15)
Wave breaking (Miche)	H/L = 0.142·tanh(kh)	Wave exists	Wave breaks
Diffraction important	D/L = 0.2	Morison equation adequate	Diffraction theory needed
Deep/shallow boundary	h/L = 0.05 (shallow); h/L = 0.5 (deep)	Shallow water: c=√(gh)	Deep water: c=gT/(2π)

C. All Force Formulas

Force	Formula	Units	Notes
Drag (steady)	½·ρ·CD·D·Uc²	N/m	Per unit length of cylinder
Drag (oscillatory, max)	½·ρ·CD·D·Um²	N/m	At time of max velocity
Inertia (max)	ρ·CM·(πD²/4)·Um·ω	N/m	At time of max acceleration
Froude-Krylov (max)	ρ·(πD²/4)·Um·ω	N/m	= FI,max / CM
FI/FD ratio	π²·CM/(KC·CD)	—	Key exam shortcut
Buoyancy	ρ·g·V	N	Upward force = displaced fluid weight
Righting moment (small ϕ)	Δ·GM_T·sin(ϕ)	N·m	Δ = displacement weight

D. All Period Formulas

```
T₁ = 2π · m₀/m₁            Tz = 2π · √(m₀/m₂)           TE = 2π · m₋₁/m₀
Tc = 2π · √(m₂/m₄)           Tp = 2π/ωp                   Hs = 4·√m₀
Encounter: ωe = ω − (ω²/g)·V·cos(β)
Heave natural: ωn = √(g/T_draught)  or  √(k/(m+m_a))
Free decay: ωn = ωd/√(1−ζ²)  where ωd = 2π/Td
```

E. Probability Formulas — Complete Set

```
P(X > x)_amplitude  = exp(−x²/(2·m₀))       [Rayleigh; peaks of narrow-band process]
P(H > h)_height     = exp(−h²/(8·m₀))        [Rayleigh for WAVE HEIGHTS: note 8m₀ not 2m₀]
Nx = (T/Tz)·exp(−x²/(2·m₀))                  [expected number of exceedances in time T]
MPL = √(2·m₀·ln N)  = σ·√(2·ln N)            [most probable largest amplitude; N=T/Tz]
H_MPL = Hs·√(0.5·ln N)                        [most probable largest wave HEIGHT]
```

```
H_rms = Hs/√2 ≈ 0.707·Hs
H_1/10 ≈ 1.27·Hs  (mean of highest 10%)
H_1/100 ≈ 1.67·Hs (mean of highest 1%)
```

CRITICAL: For AMPLITUDE exceedance use $\exp(-x^2/(2m_o))$. For HEIGHT use $\exp(-H^2/(8m_o))$. Factor of 4 difference in denominator! Height = 2×amplitude for sinusoidal waves.